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ABSTRACT

Training of general-purpose machine learning interatomic potentials (MLIPs) relies on large datasets with properties usually computed with density functional theory (DFT). A prerequisite for accurate MLIPs is that the DFT data are well converged to minimize numerical errors. A possible symptom of errors in DFT force components is nonzero net force. Here, we consider net forces in datasets including SPICE, Transition1x, ANI-1x, ANI-1xbb, AIMNet2, QCML, and OMol25. Several of these datasets suffer from significant nonzero DFT net forces. We also quantify individual force component errors by comparison to recomputed forces using more reliable DFT settings at the same level of theory, and we find significant discrepancies in force components averaging from 1.7 meV/Å in the SPICE dataset to 33.2 meV/Å in the ANI-1x dataset. These findings underscore the importance of well converged DFT data as increasingly accurate MLIP architectures become available.

I. INTRODUCTION

The development of machine learning interatomic potentials (MLIPs) has been accelerated by the availability of curated molecular datasets. Over the past decade, successively larger and more chemically complex datasets have been produced.[1–12] The developers of these datasets have built valuable foundations for general-purpose MLIPs that give access to fast and stable molecular simulations.[10,13,14]

For MLIP training, the structures are labeled with their energies and forces computed usually with density functional theory (DFT).[15,16] The quality of DFT forces is important for the accuracy of MLIPs. Unconverged electron densities and numerical errors due to approximate DFT setups can degrade the accuracy of the forces used for training and testing. Early work in the MLIP field showed improvements in force and energy prediction when less noisy data computed with tight self-consistent field settings are used.[17] While errors in forces potentially affect the quality of the MLIP, they definitely affect the test error, i.e., our understanding of how accurate the MLIP is. Considering that general-purpose MLIP force mean absolute errors and root mean square errors (RMSEs) are on the order of tens of meV/Å and sometimes down to 10 meV/Å,[12,13,18] errors in DFT forces should be much smaller, ideally 1 meV/Å and less to not affect the MLIP quality.

A clear indicator of numerical errors in the forces is the nonzero net force. The net force is obtained by taking the sum of the force components on all the atoms for each Cartesian direction and should be zero in the absence of external fields. However, errors in individual DFT force components often do not cancel, resulting in nonzero net forces. In this article, we show that the ANI-1x, Transition1x, AIMNet2, and SPICE datasets have unexpectedly large net forces, indicating suboptimal DFT settings. We then quantify the actual errors in forces by comparing to forces obtained using well converged DFT settings.

II. RESULTS

The focus of this work is on molecular datasets with available DFT forces, as listed in Table I, specifically the ANI-1xbb,[9] QCML,[11] ANI-1x[3] (small basis set 6-31G* and large basis set

TABLE I. Summary of datasets considered in this work, including their sizes in terms of the number of configurations, exchange–correlation functionals, basis sets, and DFT codes used.

Dataset	Size	Level of theory	DFT code
ANI-1xbb[9]	13.1 M	B97-3c[19]	ORCA 4[20]
QCML[11]	33.5 M	PBE0[21]	FHI-aims[22]
ANI-1x[3]	$^{(1)}$5.0 M$^{(2)}$4.6 M	ωB97x[23]/6-31G*,[24–26]$^{(2)}$def2-TZVPP[27]	$^{(1)}$Gaussian 09[28]$^{(2)}$ORCA 4
AIMNet2[10]	20.1 M	ωB97M-D3(BJ)[29–31]/def2-TZVPP	ORCA 5
Transition1x[6]	9.6 M	ωB97x/6–31G(d)[24–26]	ORCA 5.0.2
SPICE[7,8]	2.0 M	ωB97M-D3(BJ)/def2-TZVPPD[32]	Psi4[33]
OMol25[12]	100 M	ωB97M-V[29]/def2-TZVPD[32]	ORCA 6.0.0

def2-TZVPP subsets; the latter was used to train AIMNet[34]), AIMNet2,[10] Transition1x,[6] SPICE,[7,8] and OMol25[12] datasets. We note that the AIMNet2 dataset contains data from older ANI-2x[4] and OrbNet Denali[5] datasets computed with smaller basis sets.

A. Nonzero net forces reveal uncertainties in DFT calculations

Figure 1 shows the distribution of net force magnitudes divided by the number of atoms in each system for each dataset. For OMol25, we do not show the net force distribution, as we find that the net forces are exactly zero within numerical precision. From the recomputed results discussed later, we conclude that if the net force is above 1 meV/Å/atom for a given structure, then the RMSE in DFT force components tends to be above 1 meV/Å. In Fig. 1, we refer to the 1 meV/Å/atom net force boundary as the threshold. In the region above this threshold, the net force indicates the presence of significant force errors, and we have shown this region in red. Below this threshold, significant force errors may still occur, but they do not manifest as large net forces as a result of error cancellation. In particular, many configurations with net forces in the region of 0.001–1 meV/Å/atom still show force errors >1 meV/Å, and we have indicated this region in amber. Net forces below 0.001 meV/Å/atom are negligible, and we highlight this region in green. The datasets

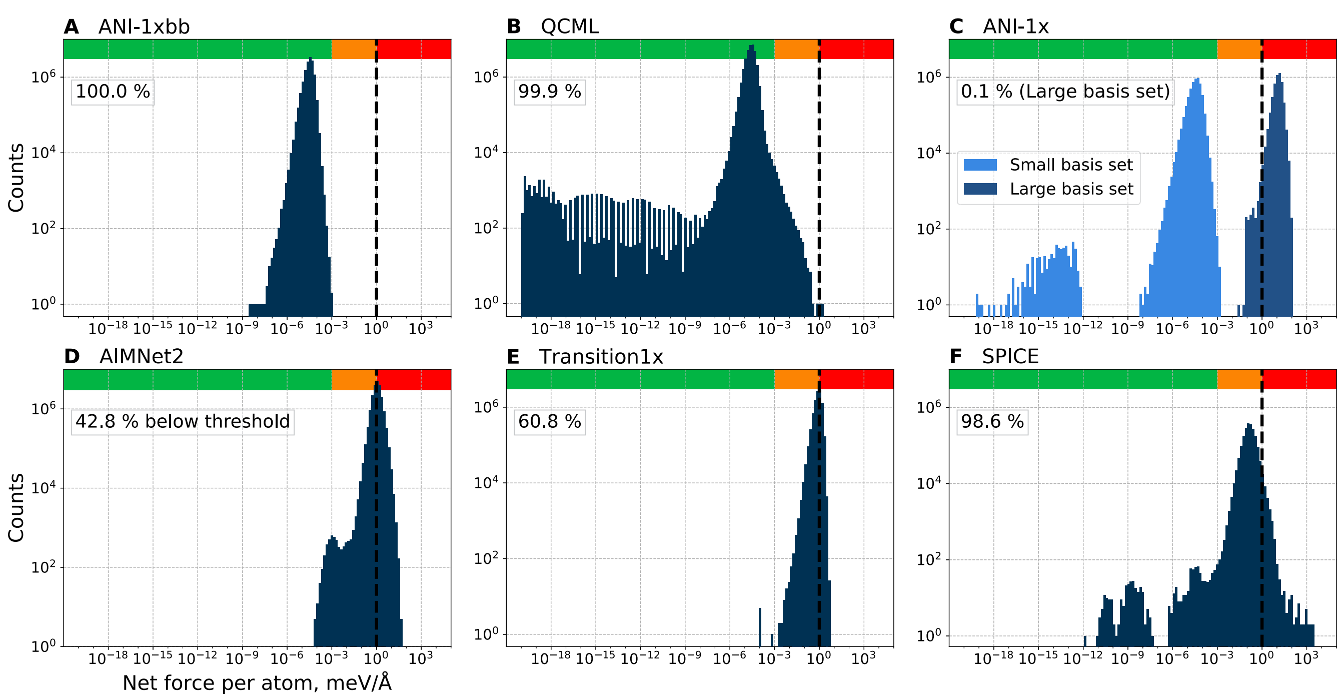

FIG. 1. Distributions of net force per atom in the ANI-1xbb, QCML, ANI-1x, AIMNet2, Transition1x, and SPICE datasets. The 1 meV/Å/atom threshold is indicated by a vertical dashed line in each plot. On the top left of each of panels **A**–**F**, the fraction of the dataset with net forces below the threshold is indicated. The red bar at the top of each panel highlights net forces above 1 meV/Å/atom, which indicate significant errors in individual force components. In the region between 10^{-3} and 1 meV/Å, indicated with an amber bar, significant DFT force errors often appear but do not result in large net forces. Negligible net forces below 10^{-3} meV/Å are in the region indicated with the green bar.

clearly differentiate into two groups: those with negligible net forces and those with net forces exceeding 1 meV/Å/atom. The ANI-1xbb, QCML, and the small basis set ANI-1x datasets fall in the first category, with most or all net forces in the green region, while in the QCML dataset, only a small fraction of structures have net forces in the amber region. In the other datasets, the net forces are much greater. In the large basis set part of ANI-1x, only 0.1% of the configurations have net forces below the threshold. Notably, the large net forces occur in the data computed with a larger basis set, intended for better quality MLIP training. In AIMNet2, Transition1x, and SPICE, 42.8%, 60.8%, and 98.6% of the data are below the 1 meV/Å/atom threshold, respectively, with most of them in the intermediate amber region. The SPICE dataset also consists of individual subsets, whose net force distributions are shown in Sec. S2 of the supplementary material.

B. Quantifying uncertainties in DFT forces

The net force acting on a system indicates numerical errors in the underlying DFT calculation. However, the net force can only be used to derive a rough estimate of the errors in the individual force components because these errors may cancel out to different degrees for different systems. To quantify the errors in individual force components arising from suboptimal DFT settings, we compare the reported forces against tightly converged reference forces computed with the same DFT functional and basis set. We note that there are also differences in total energy between different codes or settings arising mostly due to the change in absolute energy rather than noise. To compare the forces contained in the datasets with our more accurate forces, we take random samples of 1000 configurations each from the ANI-1x (large basis set), Transition1x, AIMNet2, and SPICE datasets and recompute their DFT forces using the original functional and basis set.

To identify a set of accurate computational settings, we first identify the sources of errors. We find that, in the Transition1x, ANI-1x (large basis set), and AIMNet2 datasets computed with older ORCA 4[20] and ORCA 5[35] versions, the nonzero net forces are eliminated by disabling the use of the RIJCOSX approximation.[36] This is an approximation used to accelerate the evaluation of Coulomb and exact exchange integrals. In the more recent ORCA 6.0.1 version,[37] the RIJCOSX approximation no longer causes problematic net forces and results in negligible net forces on the order of 10^{-5} meV/Å/atom. However, we still observe discrepancies in individual force components of up to several meV/Å between forces computed with and without RIJCOSX. In addition, we explore different DFT grid settings and determine that tightest grids provided by the `DEFGRID3` keyword are required, as less tight grids result in significant force deviations. Notably, ORCA 6 with `DEFGRID3` and RIJCOSX is used in the production of the OMol25 dataset. The SPICE data were computed using the Psi4 software,[33] which we compare to our recomputed ORCA forces. We also explore how to minimize the Psi4 net forces. We are able to reduce the net forces on randomly chosen SPICE systems to below 0.01 meV/Å/atom by using denser spherical and radial DFT grids, described in Sec. S1 of the supplementary material. We note that in ORCA 6.0.1, the D3 dispersion forces do not agree with the Psi4 D3 dispersion forces, which are in agreement with the forces from the standalone D3 package.[30,31] For this reason, we compare DFT forces without the D3 dispersion correction for SPICE configurations. Based on this exploration of DFT settings, we choose to use ORCA 6.0.1 using the `DEFGRID3` grid setting and with the RIJCOSX approximation disabled to obtain accurate reference forces.

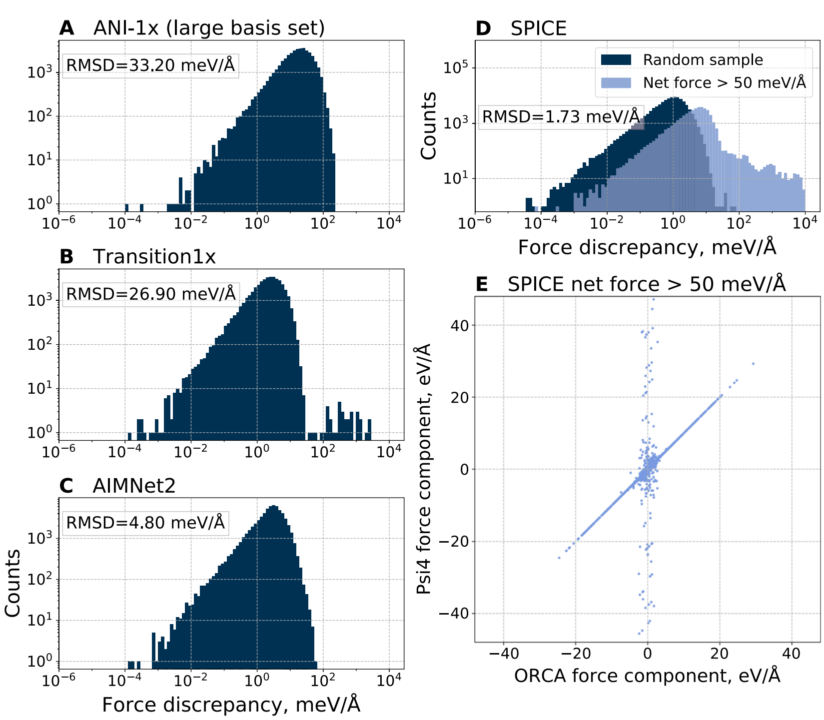

FIG. 2. Absolute discrepancies in atomic Cartesian force components between original literature values and values recomputed using ORCA 6.0.1 with our chosen settings in randomly selected structures from the ANI-1x (large basis set data), Transition1x, AIMNet2, and SPICE datasets, with their force root mean square deviation (RMSD) values, shown in panels **A**, **B**, **C**, and **D**, respectively. Each data point represents the difference in a single Cartesian force component (x, y, or z) for one atom in meV/Å. The histograms depict the distribution of these per-component discrepancies, and the RMSD values are calculated with the component-wise discrepancies. For the SPICE dataset in panel **D**, we also provide force discrepancies for all configurations whose original Psi4 net forces are greater than 50 meV/Å. Panel **E** compares original Psi4 and recomputed ORCA force components. Note that the scale in panel **E** has data in units of eV/Å, while other panels report data in meV/Å.

Figure 2 shows the absolute discrepancies between literature values and recomputed Cartesian force components for each atom individually. We show distributions of force component discrepancies for a sample of 1000 configurations for each dataset. The ANI-1x (large basis set data) and Transition1x dataset samples in panels **A** and **B** show large RMSDs of 33.2 and 26.9 meV/Å, respectively. Force discrepancy distributions show that in ANI-1x, the bulk of the data has large force discrepancies, while in Transition1x, the large force RMSD is due to a small number of outliers. Exclusion of these outliers brings the RMSD down to 4.01 meV/Å. The AIMNet2 and SPICE samples in panels **C** and **D** have force RMSD values of 4.80 meV/Å and 1.73 meV/Å, respectively, an order of magnitude lower than the other datasets. The SPICE sample has configurations from individual subsets it is composed of, which we show in detail in Sec. S2 of the supplementary material.

Species with large net forces tend to also exhibit large force errors. For illustration, we use the SPICE dataset. Since we do not obtain a representative sample of outliers due to the small sample size, we also show force discrepancies in all SPICE configurations with original Psi4 net forces greater than 50 meV/Å. As shown in panel **D**, configurations with the largest net forces also have generally greater force discrepancies compared to configurations from a random sample. Comparison of original Psi4 and recomputed ORCA force components in panel **E** shows two sets of data that follow different trends. One set is characterized by force disagreements below 10 meV/Å, which are small relative to the magnitude of the force component. The second set is characterized by force discrepancies as large as tens of thousands of meV/Å. Surprisingly, these large discrepancies occur only in force components with values smaller than 3000 meV/Å as obtained with our converged DFT settings. We find that, within a given structure, large force discrepancies of >50 meV/Å are often associated with the presence of bromine or iodine in the structure, as shown in Fig. S3 of the supplementary material.

In Fig. 3, we show force discrepancies that arise as we vary one computational setting at a time. For each sample with 1000 configurations, we calculate the RMSD in forces between different computational settings, including the DFT grid size, the use of RIJCOSX approximation, and the code version. We investigate the effect of switching on the RIJCOSX approximation while using tight DFT grids provided by the `DEFGRID3` keyword in ORCA 6.0.1. Switching on this approximation results in RMSD values below 1 meV/Å for the ANI-1x, Transition1x, and SPICE samples and 2.21 meV/Å for the AIMNet2 sample. Such errors from the RIJCOSX approximation are small on the scale of an expected MLIP force RMSE. Next, we consider how forces are affected by reducing the DFT grid sizes to the defaults provided by the `DEFGRID2` keyword. The forces change with RMSD values of 1–3 meV/Å in the ANI-1x, Transition1x, and AIMNet2 samples and by 30.1 meV/Å in the SPICE sample. The RMSD of the SPICE sample is affected by one outlier structure with force discrepancies of several eV/Å, which is why we report RMSD values with and without the outlier. Based on the ANI-1x and Transition1x sample RMSD values and the outlier in the SPICE sample, we suggest that the `DEFGRID3` tight grid settings are required for reliable forces. We also compare ORCA 6.0.1 forces to forces from older ORCA versions with less accurate DFT grids, which were used to produce the original datasets. In particular, changing to ORCA 4 for the ANI-1x sample results in an RMSD of 33.2 meV/Å. For the sample of SPICE, which was originally computed using Psi4 1.8.1 or 1.8.2, we recomputed the forces with Psi4 1.9.1 using tighter DFT grid settings to reduce nonzero net forces. The force RMSD between the forces recomputed with converged ORCA settings and recomputed Psi4 forces is 1.52 meV/Å, close to the 1 meV/Å threshold, showing consistency between the different programs if converged settings are used.

Although we focus on molecular datasets, net forces are relevant for periodic systems, too. For example, we find negligible

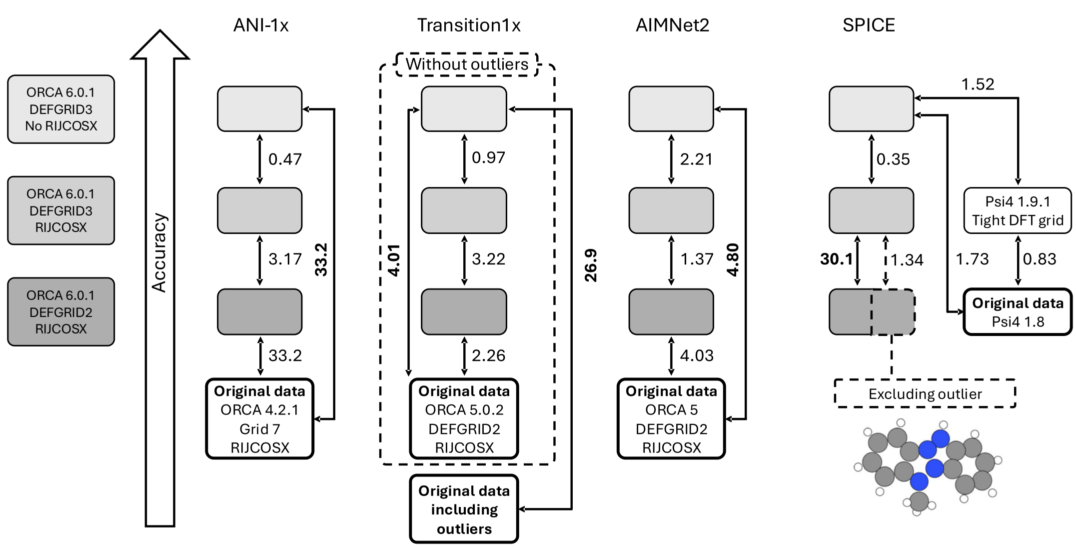

FIG. 3. Force RMSDs, in meV/Å, on the random samples containing 1000 configurations each shown in Fig. 2. Each RMSD value is shown next to an arrow connecting two boxes, and the boxes represent computational settings either shown on the left-hand side in the same shade of gray or written explicitly in the box. For Transition1x, we also show the effect of retaining outliers on force RMSD. For SPICE, we show the single outlier that alone is responsible for changing RMSD from 1.34 to 30.1 meV/Å.

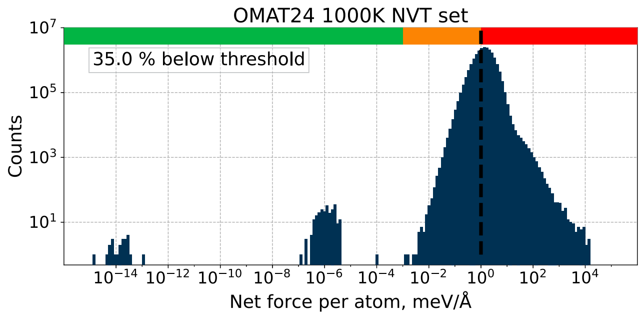

FIG. 4. Distribution of per atom net forces in the 1000 K NVT subset of OMAT24. 65% of the configurations exceed 1 meV/Å/atom net force threshold, with thousands even exceeding 100 meV/Å/atom. The red bar on top indicates the region above the 1 meV/Å/atom threshold. The green bar indicates negligible net forces below 10^{-3} meV/Å/atom, and the amber bar indicates net forces of intermediate size.

net forces of less than 10^{-5} meV/Å/atom in a 10% subset of the Materials Project Trajectory (MPTrj)[38] dataset. In addition, we find significant nonzero DFT net forces in the 1000 K NVT subset from the OMAT24[39] materials dataset, with thousands of configurations exhibiting net forces exceeding 100 meV/Å/atom, as shown in Fig. 4. We find that the nonzero force problem is solved by using a tighter energy convergence threshold specified by the `EDIFF` keyword in VASP,[40] concluding that the threshold of 5×10^{-5} eV per atom used in OMAT24 is too loose. An example structure from OMAT24 computed with different energy convergence thresholds is given in Sec. S4 of the supplementary material. In the same section, we also collect net forces for other periodic systems computed with VASP and other popular periodic DFT codes, including CASTEP,[41] CP2K,[42] and FHI-aims.[22]

III. DISCUSSION

To reiterate, DFT forces are included in molecular datasets with the primary goal to provide training data for MLIPs. Therefore, we draw the reader's attention to the potential effect of the demonstrated force discrepancies on the quality of an MLIP trained using these data. We may consider force discrepancies due to numerical precision of DFT as a contribution to the overall MLIP error. In principle, an MLIP may be able to cope with noisy DFT training data, as it has been shown with models trained using the Open Catalyst 2025 dataset.[43] However, it is difficult to assess the accuracy of the MLIP using test data that are noisy as well. Therefore, errors in DFT forces should ideally be much smaller than the expected MLIP force errors. Since general-purpose MLIP force errors can be as low as 10 meV/Å, the DFT force RMSD of 4.8 meV/Å, as in the AIMNet2 dataset sample, is of the size where DFT force discrepancies start to affect the accuracy of the MLIP. On the other hand, force RMSDs of around 30 meV/Å, as in ANI-1x and in Transition1x without the removal of outliers, may significantly limit the overall accuracy of the models trained on these data.

IV. CONCLUSIONS

To conclude, large DFT datasets are incredibly valuable for the modeling of molecules and materials. However, we have shown that the DFT data of numerous molecular datasets suffer from unphysical nonzero net forces, which are a symptom of numerical problems in the DFT calculation. At times, the force discrepancies are significant, reaching or exceeding the force errors expected from state-of-the-art general-purpose MLIPs. We have established settings that allow us to obtain reliable DFT reference forces and used them to quantify the error in the DFT forces by computing reliable reference forces for fractions of those datasets. These results lead to important guidelines for considering existing datasets and the development of new datasets. For existing datasets, care must be taken regarding the quality of the DFT data. Low quality DFT forces can be addressed to some extent by filtering. For the development of new datasets, good quality DFT settings should be chosen to minimize numerical errors and the associated noise. Net forces are an indicator of force errors, and we recommend that more attention is paid to them. As new datasets of molecules and materials are being developed, and MLIP architectures are able to fit reference data with increasing accuracy, we emphasize that labeling databases with high-quality DFT data is increasingly important.

SUPPLEMENTARY MATERIAL

See the supplementary material for computational details, overview of net forces and force discrepancies in the SPICE dataset subsets, examples of molecules in the SPICE dataset with large DFT force errors, and additional examples of net force data from periodic DFT calculations.

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AUTHOR DECLARATIONS

Conflict of Interest

G.C. has equity interest in Symmetric Group LLP that licenses force fields commercially, and in Ångstrom AI.

Author Contributions

Domantas Kuryla: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Fabian Berger**: Conceptualization (equal); Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Gábor Csányi**: Conceptualization (equal); Formal analysis (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Angelos Michaelides**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Computed net forces for all datasets discussed in this work and recomputed DFT forces for random data samples are available on GitHub at water-ice-group/datasets_dft_accuracy.

REFERENCES

[1]R. Ramakrishnan, P. O. Dral, M. Rupp, and O. A. von Lilienfeld, "Quantum chemistry structures and properties of 134 kilo molecules," Sci. Data **1**, 140022 (2014).

[2]S. Chmiela, A. Tkatchenko, H. E. Sauceda, I. Poltavsky, K. T. Schütt, and K.-R. Müller, "Machine learning of accurate energy-conserving molecular force fields," Sci. Adv. **3**, e1603015 (2017).

[3]J. S. Smith, R. Zubatyuk, B. Nebgen, N. Lubbers, K. Barros, A. E. Roitberg, O. Isayev, and S. Tretiak, "The ANI-1ccx and ANI-1x data sets, coupled-cluster and density functional theory properties for molecules," Sci. Data **7**, 134 (2020).

[4]C. Devereux, J. S. Smith, K. K. Huddleston, K. Barros, R. Zubatyuk, O. Isayev, and A. E. Roitberg, "Extending the applicability of the ANI deep learning molecular potential to sulfur and halogens," J. Chem. Theory Comput. **16**, 4192–4202 (2020).

[5]A. S. Christensen, S. K. Sirumalla, Z. Qiao, M. B. O'Connor, D. G. A. Smith, F. Ding, P. J. Bygrave, A. Anandkumar, M. Welborn, F. R. Manby, and T. F. Miller, "OrbNet Denali: A machine learning potential for biological and organic chemistry with semi-empirical cost and DFT accuracy," J. Chem. Phys. **155**, 204103 (2021).

[6]M. Schreiner, A. Bhowmik, T. Vegge, J. Busk, and O. Winther, "Transition1x—A dataset for building generalizable reactive machine learning potentials," Sci. Data **9**, 779 (2022).

[7]P. Eastman, P. K. Behara, D. L. Dotson, R. Galvelis, J. E. Herr, J. T. Horton, Y. Mao, J. D. Chodera, B. P. Pritchard, Y. Wang, G. De Fabritiis, and T. E. Markland, "SPICE, a dataset of drug-like molecules and peptides for training machine learning potentials," Sci. Data **10**, 11 (2023).

[8]P. Eastman, B. P. Pritchard, J. D. Chodera, and T. E. Markland, "Nutmeg and SPICE: Models and data for biomolecular machine learning," J. Chem. Theory Comput. **20**, 8583–8593 (2024).

[9]S. Zhang, R. Zubatyuk, Y. Yang, A. Roitberg, and O. Isayev, "ANI-1xBB: An ANI-based reactive potential for small organic molecules," J. Chem. Theory Comput. **21**, 4365–4374 (2025).

[10]D. M. Anstine, R. Zubatyuk, and O. Isayev, "AIMNet2: A neural network potential to meet your neutral, charged, organic, and elemental-organic needs," Chem. Sci. **16**, 10228–10244 (2025).

[11]S. Ganscha, O. T. Unke, D. Ahlin, H. Maennel, S. Kashubin, and K.-R. Müller, "The qcml dataset, quantum chemistry reference data from 33.5 m DFT and 14.7b semi-empirical calculations," Sci. Data **12**, 406 (2025).

[12]D. S. Levine, M. Shuaibi, E. W. C. Spotte-Smith, M. G. Taylor, M. R. Hasyim, K. Michel, I. Batatia, G. Csányi, M. Dzamba, P. Eastman, N. C. Frey, X. Fu, V. Gharakhanyan, A. S. Krishnapriyan, J. A. Rackers, S. Raja, A. Rizvi, A. S. Rosen, Z. Ulissi, S. Vargas, C. L. Zitnick, S. M. Blau, and B. M. Wood, "The open molecules 2025 (OMol25) dataset, evaluations, and models," arXiv:2505.08762 [physics.chem-ph] (2025).

[13]D. P. Kovács, J. H. Moore, N. J. Browning, I. Batatia, J. T. Horton, Y. Pu, V. Kapil, W. C. Witt, I.-B. Magdău, D. J. Cole, and G. Csányi, "Mace-off: Short-range transferable machine learning force fields for organic molecules," J. Am. Chem. Soc. **147**, 17598–17611 (2025).

[14]J. H. Moore, D. J. Cole, and G. Csanyi, "Computing hydration free energies of small molecules with first principles accuracy," arXiv:2405.18171 [physics.chem-ph] (2024).

[15]P. Hohenberg and W. Kohn, "Inhomogeneous electron gas," Phys. Rev. **136**, B864–B871 (1964).

[16]W. Kohn and L. J. Sham, "Self-consistent equations including exchange and correlation effects," Phys. Rev. **140**, A1133–A1138 (1965).

[17]A. S. Christensen and O. A. Von Lilienfeld, "On the role of gradients for machine learning of molecular energies and forces," Mach. Learn.: Sci. Technol. **1**, 045018 (2020).

[18]I. Batatia, P. Benner, Y. Chiang, A. M. Elena, D. P. Kovács, J. Riebesell, X. R. Advincula, M. Asta, M. Avaylon, W. J. Baldwin, F. Berger, N. Bernstein, A. Bhowmik, S. M. Blau, V. Cărare, J. P. Darby, S. De, F. D. Pia, V. L. Deringer, R. Elijošius, Z. El-Machachi, F. Falcioni, E. Fako, A. C. Ferrari, A. Genreith-Schriever, J. George, R. E. A. Goodall, C. P. Grey, P. Grigorev, S. Han, W. Handley, H. H. Heenen, K. Hermansson, C. Holm, J. Jaafar, S. Hofmann, K. S. Jakob, H. Jung, V. Kapil, A. D. Kaplan, N. Karimitari, J. R. Kermode, N. Kroupa, J. Kullgren, M. C. Kuner, D. Kuryla, G. Liepuoniute, J. T. Margraf, I.-B. Magdău, A. Michaelides, J. H. Moore, A. A. Naik, S. P. Niblett, S. W. Norwood, N. O'Neill, C. Ortner, K. A. Persson, K. Reuter, A. S. Rosen, L. L. Schaaf, C. Schran, B. X. Shi, E. Sivonxay, T. K. Stenczel, V. Svahn, C. Sutton, T. D. Swinburne, J. Tilly, C. van der Oord, E. Varga-Umbrich, T. Vegge, M. Vondrák, Y. Wang, W. C. Witt, F. Zills, and G. Csányi, "A foundation model for atomistic materials chemistry," J. Chem. Phys. **163**, 184110 (2025).

[19]J. G. Brandenburg, C. Bannwarth, A. Hansen, and S. Grimme, "B97-3c: A revised low-cost variant of the B97-D density functional method," J. Chem. Phys. **148**, 064104 (2018).

[20]F. Neese, F. Wennmohs, U. Becker, and C. Riplinger, "The orca quantum chemistry program package," J. Chem. Phys. **152**, 224108 (2020).

[21]C. Adamo and V. Barone, "Toward reliable density functional methods without adjustable parameters: The PBE0 model," J. Chem. Phys. **110**, 6158–6170 (1999).

[22]V. Blum, R. Gehrke, F. Hanke, P. Havu, V. Havu, X. Ren, K. Reuter, and M. Scheffler, "*Ab initio* molecular simulations with numeric atom-centered orbitals," Comput. Phys. Commun. **180**, 2175–2196 (2009).

[23]J.-D. Chai and M. Head-Gordon, "Systematic optimization of long-range corrected hybrid density functionals," J. Chem. Phys. **128**, 084106 (2008).

[24]R. Ditchfield, W. J. Hehre, and J. A. Pople, "Self-consistent molecular-orbital methods. IX. An extended gaussian-type basis for molecular-orbital studies of organic molecules," J. Chem. Phys. **54**, 724–728 (1971).

[25]W. J. Hehre, R. Ditchfield, and J. A. Pople, "Self—Consistent molecular orbital methods. XII. Further extensions of Gaussian—type basis sets for use in molecular orbital studies of organic molecules," J. Chem. Phys. **56**, 2257–2261 (1972).

[26]P. C. Hariharan and J. A. Pople, "The influence of polarization functions on molecular orbital hydrogenation energies," Theor. Chim. Acta **28**, 213–222 (1973).

[27]F. Weigend and R. Ahlrichs, "Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy," Phys. Chem. Chem. Phys. **7**, 3297–3305 (2005).

[28]M. Frisch, G. Trucks, H. Schlegel, G. Scuseria, M. Robb, J. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. Hratchian, A. Izmaylov, J. Bloino, G. Zheng, J. Sonnenberg, M. Hada, and D. Fox, *Gaussian 09 Revision a.1*, Gaussian, Inc., Wallingford, CT, 2009.

[29]N. Mardirossian and M. Head-Gordon, "ωB97M-V: A combinatorially optimized, range-separated hybrid, meta-GGA density functional with VV10 nonlocal correlation," J. Chem. Phys. **144**, 214110 (2016).

[30]S. Grimme, J. Antony, S. Ehrlich, and H. Krieg, "A consistent and accurate *ab initio* parametrization of density functional dispersion correction (DFT-d) for the 94 elements H-Pu," J. Chem. Phys. **132**, 154104 (2010).

[31]S. Grimme, S. Ehrlich, and L. Goerigk, "Effect of the damping function in dispersion corrected density functional theory," J. Comput. Chem. **32**, 1456–1465 (2011).

[32] D. Rappoport and F. Furche, "Property-optimized Gaussian basis sets for molecular response calculations," J. Chem. Phys. **133**, 134105 (2010).
[33] D. G. A. Smith, L. A. Burns, A. C. Simmonett, R. M. Parrish, M. C. Schieber, R. Galvelis, P. Kraus, H. Kruse, R. Di Remigio, A. Alenaizan, A. M. James, S. Lehtola, J. P. Misiewicz, M. Scheurer, R. A. Shaw, J. B. Schriber, Y. Xie, Z. L. Glick, D. A. Sirianni, J. S. O'Brien, J. M. Waldrop, A. Kumar, E. G. Hohenstein, B. P. Pritchard, B. R. Brooks, H. F. Schaefer, A. Y. Sokolov, K. Patkowski, A. E. DePrince, U. Bozkaya, R. A. King, F. A. Evangelista, J. M. Turney, T. D. Crawford, and C. D. Sherrill, "PSI4 1.4: Open-source software for high-throughput quantum chemistry," J. Chem. Phys. **152**, 184108 (2020).
[34] R. Zubatyuk, J. S. Smith, J. Leszczynski, and O. Isayev, "Accurate and transferable multitask prediction of chemical properties with an atoms-in-molecules neural network," Sci. Adv. **5**, eaav6490 (2019).
[35] F. Neese, "Software update: The orca program system—Version 5.0," WIREs Comput. Mol. Sci. **12**, e1606 (2022).
[36] B. Helmich-Paris, B. De Souza, F. Neese, and R. Izsák, "An improved chain of spheres for exchange algorithm," J. Chem. Phys. **155**, 104109 (2021).
[37] F. Neese, "Software update: The orca program system—Version 6.0," WIREs Comput. Mol. Sci. **15**, e70019 (2025).
[38] B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, "CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling," Nat. Mach. Intell. **5**, 1031–1041 (2023).
[39] L. Barroso-Luque, M. Shuaibi, X. Fu, B. M. Wood, M. Dzamba, M. Gao, A. Rizvi, C. L. Zitnick, and Z. W. Ulissi, "Open materials 2024 (OMat24) inorganic materials dataset and models," arXiv:2410.12771 [cond-mat.mtrl-sci] (2024).
[40] G. Kresse and J. Hafner, "*Ab initio* molecular dynamics for liquid metals," Phys. Rev. B **47**, 558–561 (1993).
[41] S. J. Clark, M. D. Segall, C. J. Pickard, P. J. Hasnip, M. I. J. Probert, K. Refson, and M. C. Payne, "First principles methods using castep," Z. Kristallogr.-Crystalline Mater. **220**, 567–570 (2005).
[42] T. D. Kühne, M. Iannuzzi, M. Del Ben, V. V. Rybkin, P. Seewald, F. Stein, T. Laino, R. Z. Khaliullin, O. Schütt, F. Schiffmann, D. Golze, J. Wilhelm, S. Chulkov, M. H. Bani-Hashemian, V. Weber, U. Borštnik, M. Taillefumier, A. S. Jakobovits, A. Lazzaro, H. Pabst, T. Müller, R. Schade, M. Guidon, S. Andermatt, N. Holmberg, G. K. Schenter, A. Hehn, A. Bussy, F. Belleflamme, G. Tabacchi, A. Glöß, M. Lass, I. Bethune, C. J. Mundy, C. Plessl, M. Watkins, J. VandeVondele, M. Krack, and J. Hutter, "CP2k: An electronic structure and molecular dynamics software package—Quickstep: Efficient and accurate electronic structure calculations," J. Chem. Phys. **152**, 194103 (2020).
[43] S. J. Sahoo, M. Maraschin, D. S. Levine, Z. Ulissi, C. L. Zitnick, J. B. Varley, J. A. Gauthier, N. Govindarajan, and M. Shuaibi, "The open catalyst 2025 (OC25) dataset and models for solid-liquid interfaces," arXiv:2509.17862 [cond-mat.mtrl-sci] (2025).